Comp 3016 CW1 Proposal

Game Title

The Scarygon Trail

Genre

The story of the game is generative, combining comic horror and survival resources management storytelling, inspired by The Oregon Trail.

Synopsis

Your wagon company is going toward the west in the early 1800s. Nothing in that sentence has prepared you for everything that will happen to you after that journey begins. Each stop along the road tells us a brief narrative incident – with the same tone, whether it is a broken wheel or your son's attempt to defy Christianity and try to place a hex on you. The dread is genuine, the stakes are sharp, but the game never looks at you in the eye, it is up to your son to do that.

Core Gameplay Loop

1. Proceed to the next location along the trail (SDL3-produced wagon in motion on the map).
2. Check status - the upper side indicates information about the group members, inventory and condition of the wagon in a glance.
3. Solve a situation - on the bottom side, narrative text is displayed as well as 2–4 other buttons.
4. Lay down the consequences - based on the decisions made, either resources, sanity or group members will change.
5. Do this until the journey comes to an end or the group is defeated.

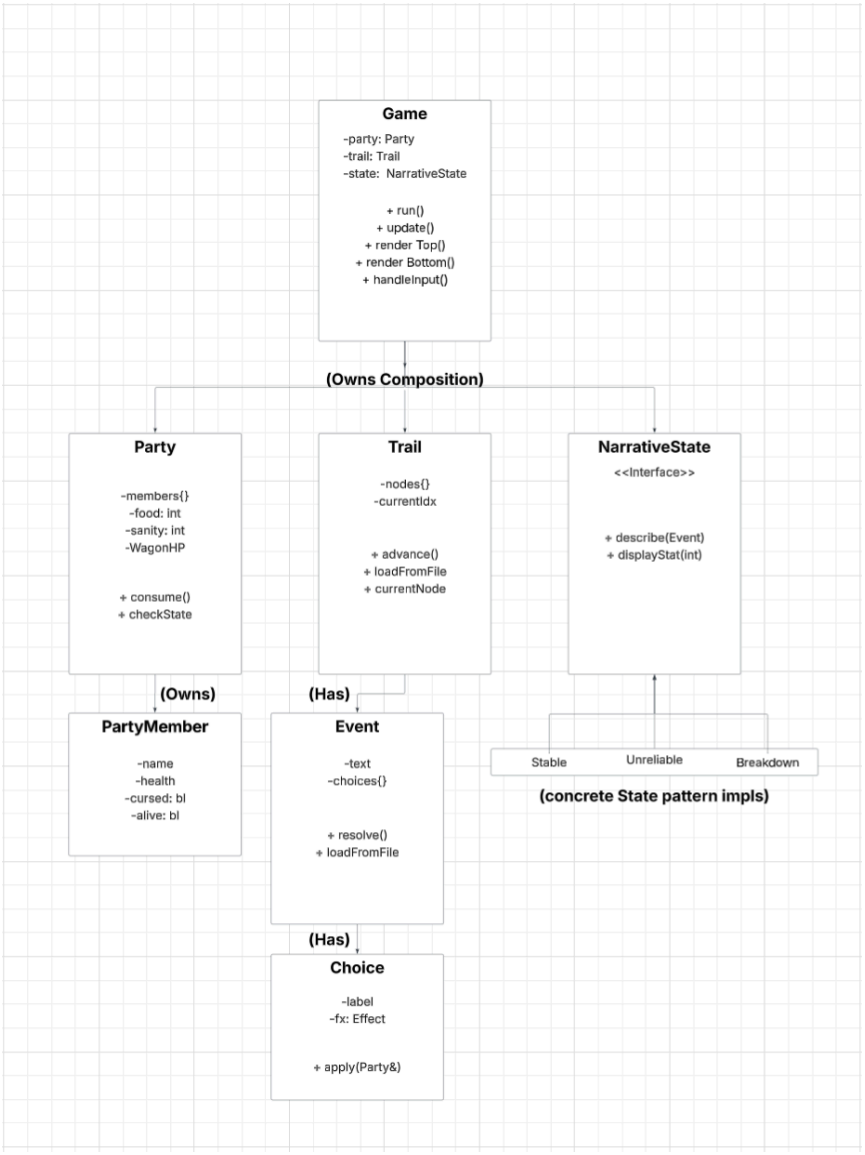
Intended Mechanics

- Management of resources includes food, health, sanity, and condition of the wagon, which erodes as it travels down the trail.
- Events encountered vary from mundane (such as running low on supplies or having a sick ox) to bizarre (such as a cursed heirloom or cults in the woods).
- Losing your grip on sanity will make the game unreliable in plotting events — in other words, the less you are sane, the less reliable game becomes, mixing black humor with horror.
- Different outcomes (in terms of risk/reward) for making decisions in each event (e.g., attempt to fix the wagon wheel yourself or trust a stranger who offers you help).
- Each character in the party can have an independent health status (sick, cursed, lost, dead) rather than being a shared pool.

Intended Technical Features

- The application utilizes C++ with SDL3 in order to perform rendering and input-related tasks. The user interface (UI) of the application consists of two viewports where the upper part deals with the graphical environment (the wagon, the characters sprite, icons of inventory) while the bottom view shows the scrolling text and registers button input.
- All events and the text are stored on local files. This means that when the application is launched, the relevant text is fetched directly from the files.
- Thus, the principles of object-oriented programming and design patterns are used in the application. In particular, there are several classes including Party, PartyMember, TrailNode, Event, and Choice.
- The state pattern will be employed to differentiate between the stable (reliable) narrative and the unstable narrative.
- The usage of the observer pattern will allow for events to occur depending upon the state of resources (e.g. if the amount of food is small, the game will trigger events leading to irreversible decisions).
- The input validation ensures that no corrupted file can lead to application freezing.

UML Sketch



Architecture Overview

- The main loop of the game is controlled by the Game class that interacts with the rendering processes (the SDL3 visual screen in front of the user and the text/choices screen). In turn, the Trail class emulates the array of TrailNode objects where each node corresponds to an appropriate Event read from a file.
- The Event consists of a text and an array of Choice objects, each of which has its own effects on the resources and sanity of a character depending on the choice.
- The Party is created by several PartyMember objects and can generate and destroy its members (referring to the Party's composition).
- The NarrativeState class is built according to the principle of the State pattern enabling the event text to be represented in a more or less reliable manner as for the resources quantity.

External Libraries

- SDL3 (mandatory– rendering, window, input)
- No other libraries planned at the moment; if looking for new libraries later (for instance, for sound work, it may be irrKlang), it will be applied for approval.

Intended Use of AI

ChatGPT will be used for:

- Drafting and refining the text of absurd/comedic events and its outcomes
- Creating ideas for the table of increasing events
- Writing some parts of code to reduce time
- Debugging and review support
- Help with the structure of documentation or reports

According to what was said in the brief, all applications will be provided with the AI Development Log recording the necessary tools, evidence, classification of use, and ways for verification.

Development Roadmap

Week	Milestone
1	Planning completion of 30% and finishing proposal document.
2	Proposal sent/split-panel skeleton and SDL3 windows created.
3	Trail/Party/Event classes created/file loading for events/initial git commits/Working of travel loop/core rendering panel made.
4	Decision system has consequences/sanity mechanic/unreliable narration.
5	Complete event table/playtest round 1 done (fixing bugs).
6	Refining/balancing completion of AI/report/video presented.